Comparative Analysis of ML Models for Accurate Real Estate Valuation in Dubai

Khadejeh Alhadhoud
School of Computer Science
University of Wollongong Dubai
Dubai, United Arab Emirates
Kirah733@uowmail.edu.au

Marina Elkedwany
School of Computer Science
University of Wollongong Dubai
Dubai, United Arab Emirates
mse636@uowmail.edu.au

Laila Mohamed Elsayed
School of Computer Science
University of Wollongong Dubai
Dubai, United Arab Emirates
lme131@uowmail.edu.au

***Abstract*—Dubai's real estate market includes many neighborhoods and property types, which makes price prediction challenging. Prices are influenced by location, size, and property characteristics, so simple linear assumptions do not always capture the patterns in real transaction data. This paper evaluates whether tree-based models provide better valuation accuracy than linear baselines on a large Dubai Land Department (DLD) transactions dataset.**

We used 469,409 DLD sales transactions. After cleaning and basic exploratory analysis, we trained two regression models (Linear Regression and Random Forest) to predict the log-transformed transaction price, and two classification models (Logistic Regression and Decision Tree) to classify properties into three price groups (Low, Medium, and High). On the test set, Random Forest improves R^2 from 0.7663 to 0.8950 (a 16.79% increase) and reduces prediction error (MAE 48.35%, RMSE 32.96% reduction) compared to Linear Regression. For classification, Decision Tree achieves 0.8843 accuracy/F1 compared to 0.8113 for Logistic Regression (an improvement of 7.30 percentage points). Overall, Random Forest and Decision Tree perform better on this dataset because they capture nonlinear relationships and feature interactions common in real estate pricing, especially size and location.

***Index Terms*—Real Estate, Machine Learning, Regression, Classification, Dubai Land Department, Prediction**

I. Introduction

Real estate prices in Dubai depend on many factors, including property size, neighborhood, property type, and transaction timing. These factors do not always have a simple linear relationship with price, and the market includes a mix of new developments, luxury areas, and different building styles. Because of this, predicting prices is not straightforward, and large errors can be costly for buyers, sellers, and investors.

This paper uses a large transactions dataset from the Dubai Land Department (DLD) and focuses on sales transactions. We apply data cleaning, feature encoding, and a consistent evaluation setup to compare a linear baseline with a tree-based model for regression, and to compare two standard classification models for price grouping.

The main finding is that tree-based models consistently outperform the linear baseline on this dataset. For example, Random Forest improves test R^2 from 0.7663 to 0.8950 (an increase of 16.79%), and Decision Tree improves classification performance from 0.8113 to 0.8843 (about 7 percentage points) compared to Logistic Regression.

Contributions: (1) We present a practical workflow for cleaning DLD transactions, encoding mixed feature types, and evaluating models fairly. (2) We show the performance difference between linear baselines and tree-based methods for both regression and price-segment classification on a large Dubai dataset. (3) We report 5-fold cross-validation results with low standard deviations to show that the findings are stable across folds.

II. Related Work

Machine learning for property valuation is commonly treated as a supervised learning problem where a model predicts the sale price based on property characteristics. Traditional hedonic approaches often assume linear relationships, but real estate data can be nonlinear and influenced by interacting factors such as location and property type. [1]

Tree-based methods are widely used for property valuation because they can capture nonlinear patterns and interactions. Random Forest builds many decision trees and averages their predictions, which can improve accuracy and reduce overfitting compared to a single tree. [2]

Recent studies in Dubai/UAE real estate also report that ensemble or tree-based models can outperform linear baselines, especially when the dataset contains mixed property types and strong location effects. [3]–[5]

Our work follows this direction by evaluating both regression and price-group classification on a large DLD transactions dataset and by adding interpretability (SHAP) to explain which features push predictions up or down.

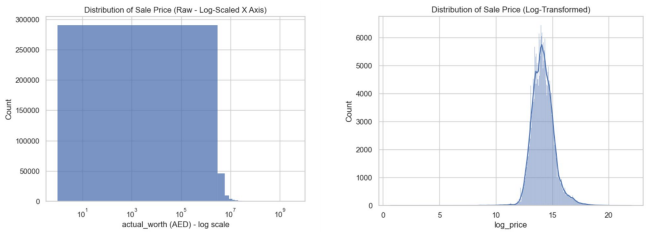
III. Dataset and Preprocessing

We used a cleaned Dubai Land Department transactions file with 469,409 sales records and 32 columns. The transaction price distribution is strongly skewed, so we applied a logarithmic transformation, $\log(1+\text{price})$, to reduce the impact of extreme values and make the target distribution more balanced.

For cleaning, we kept only sales transactions and removed rows with missing price or missing area. We also converted the transaction date into year and month. The rooms field was cleaned by extracting numeric values and treating Studio as 0. Categorical variables were one-hot encoded, and numeric features were passed through.

TABLE I: Main features used in the models.

Feature	Type
procedure_area	Numeric
rooms	Numeric
year, month	Numeric
property_type_en	Categorical
property_usage_en	Categorical
area_name_en	Categorical
reg_type_en	Categorical



(a) Raw price distribution (log-scale x-axis).

(b) Log-transformed price distribution.

Fig. 1: Price distribution before and after applying $\log(1 + \text{price})$.

TABLE II: Top six areas by number of transactions.

Area	Count
Marsa Dubai	30041
Al Barsha South Fourth	25143
Business Bay	21727
Al Thanyah Fifth	20557
Burj Khalifa	20144
Al Warsan First	14680

IV. Feature Description

The dataset contains both numeric and categorical features. Numeric features include procedure_area, rooms, and date-based features (year and month). Categorical features include property_type_en, property_usage_en, area_name_en, and reg_type_en. We use one-hot encoding for categorical features and keep numeric features as they are (with scaling for linear models). [6]

V. Exploratory Data Analysis

Figure 1 shows the effect of the logarithmic transformation. Raw prices exhibit a long right tail, reflecting a small number of extremely expensive properties, but after the transformation the distribution becomes much closer to normal.

To check location frequency, we counted how often each area appears in the dataset. Table II lists the top six areas by number of transactions.

VI. Experimental Setup

We split the data into 80% training and 20% testing, using a fixed random seed (random_state = 42) so the results are consistent.

TABLE III: 5-fold cross-validation results.

Model	Metric	Mean	Std
Linear Regression	R^2	0.773887	0.004085
Random Forest	R^2	0.902418	0.003857
Logistic Regression	Accuracy	0.810963	0.001965
Decision Tree	Accuracy	0.884168	0.001135

Categorical features are one-hot encoded with handle_unknown='ignore'. For regression, we use a pipeline with preprocessing plus the model. [6]

We keep the model setups simple: Random Forest Regressor uses 100 trees (n_estimators=100), with max_depth=None and n_jobs=-1. Linear Regression is used as the baseline. [2]

To make results more reliable than a single split, we also performed 5-fold cross-validation (Table III). We used KFold for regression and StratifiedKFold for classification. [7]

The standard deviations across folds are very small (around 0.004 for regression R^2 and around 0.002 for classification accuracy), indicating that model performance is consistent across different data splits. This suggests that the observed performance improvements are robust rather than caused by a particular train–test split.

VII. Methodology

For regression, we predict the log-transformed transaction price and evaluate using MAE, RMSE, and R^2.

For classification, we create three price groups using fixed thresholds on the original price: Low ($<$ 1,000,000 AED), Medium (1,000,000 to 3,000,000 AED), and High ($\geq$ 3,000,000 AED). We then train Logistic Regression and Decision Tree classifiers and evaluate accuracy, precision, recall, F1 score, and confusion matrix. [8], [9]

VIII. Discussion

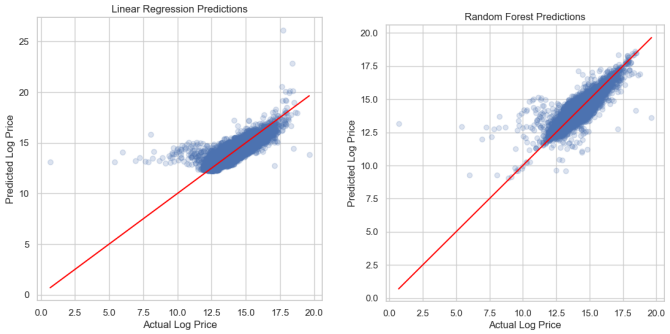
Table IV shows the regression results on the test set. Random Forest has lower errors and explains more variance than Linear Regression.

Figure 2 compares predicted vs. actual log prices for both models. Most points are close to the diagonal, meaning the models predict well across typical price ranges. Random Forest shows tighter alignment, especially in the dense middle range of prices.

The cross-validation standard deviations in Table III are very small, which suggests the results are stable and not caused by one lucky train-test split.

For classification, the Decision Tree achieves higher accuracy than Logistic Regression (Table V). The confusion matrix (Figure 3) shows better separation between the three price categories.

Decision Tree feature importance (Figure 4) suggests procedure_area is the strongest factor, with time (year/month) and rooms/area also contributing. This makes sense because, in real estate, size and location-related information usually have the biggest effect on price.



(a) Linear Regression. (b) Random Forest.

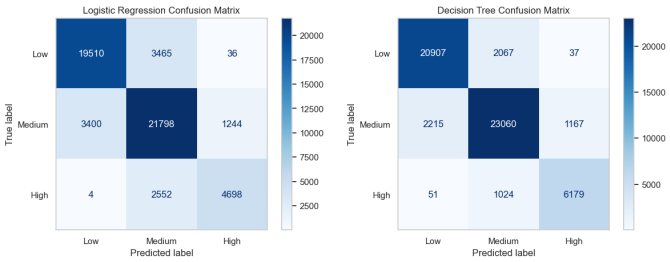
Fig. 2: Predicted vs. actual log prices for Linear Regression and Random Forest.

TABLE IV: Regression results on the test set.

Model	MAE	RMSE	R^2
Linear Regression	0.269555	0.389227	0.766280
Random Forest	0.139233	0.260947	0.894950

TABLE V: Classification results.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.811293	0.81	0.81	0.81
Decision Tree	0.884300	0.88	0.88	0.88



(a) Logistic Regression. (b) Decision Tree.

Fig. 3: Confusion matrices for Logistic Regression and Decision Tree.

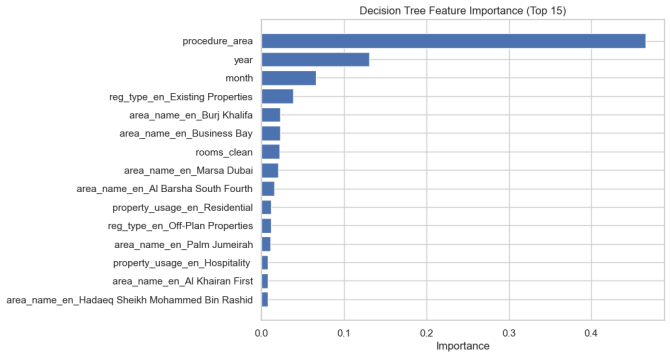


Fig. 4: Top 15 most important features from the Decision Tree model.

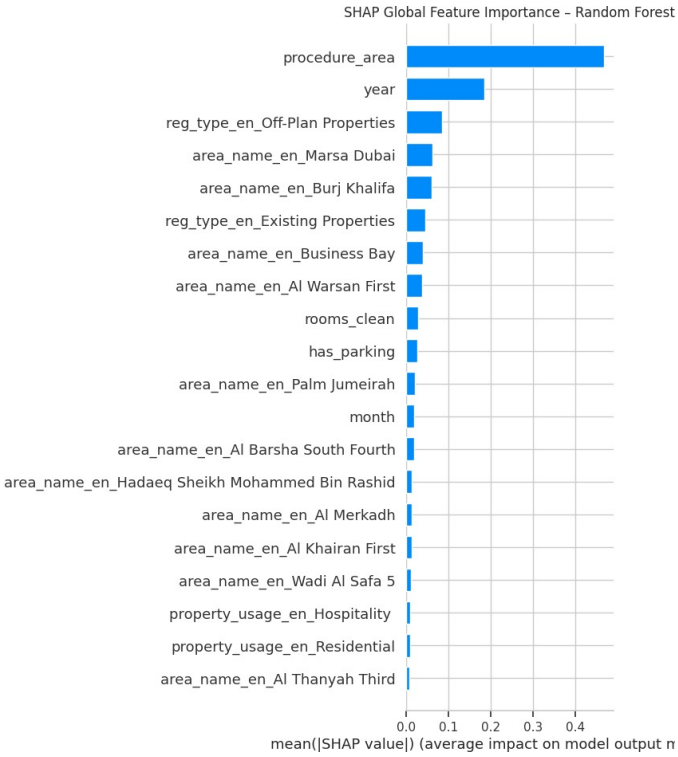


Fig. 5: SHAP global feature importance (mean absolute SHAP values).

IX. Model Interpretability (SHAP)

To better understand what the Random Forest is learning, we used SHAP (SHapley Additive exPlanations). SHAP provides a contribution value for each feature in each predicted log price, indicating whether it pushes the prediction upward or downward, which helps explain why the model predicts a higher or lower price.

The SHAP global feature importance shows that *procedure_area* is the most influential feature, which is expected because size strongly affects property price. Temporal effects (especially *year*) also matter, likely reflecting changes in market conditions over time. Some location indicators (e.g., Marsa Dubai and Burj Khalifa) appear among the top drivers, which matches the idea that pricing differs by neighborhood. The summary plot shows directionality: higher values of *procedure_area* generally increase predicted log price, while lower values decrease it.

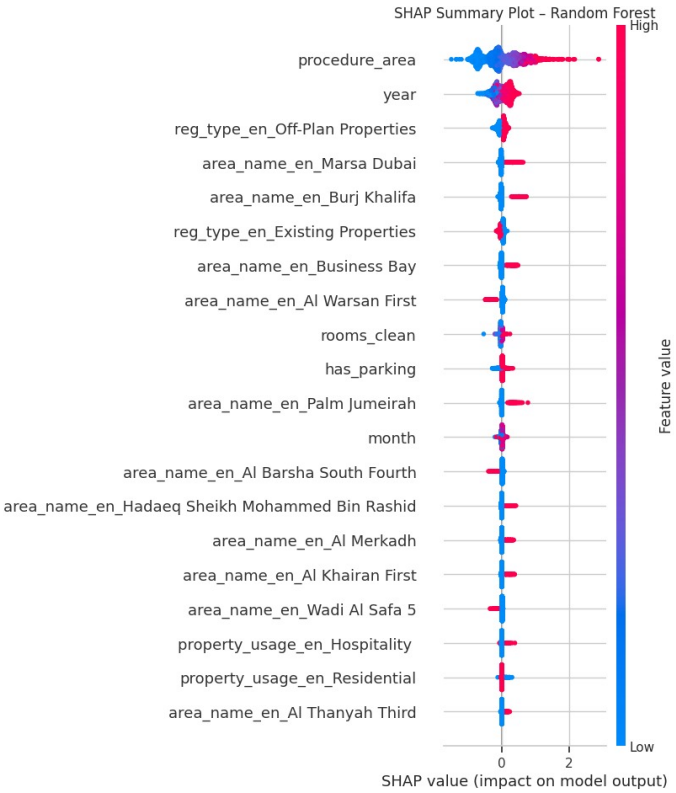


Fig. 6: SHAP summary plot (direction of feature impact on log price).

X. Error Analysis

For regression, errors are larger for very expensive properties compared to typical price ranges. Even with the log transform, rare high-value transactions remain harder to predict because the model sees fewer similar examples during training. Also, luxury properties may include factors not captured in the dataset (for example: view, finishing quality, or facilities), which can increase error.

For classification, most mistakes happen near the boundaries between the Low and Medium price groups and the Medium and High price groups. This is expected because properties near the thresholds can have very similar features, which makes them harder to separate.

XI. Limitations

This work uses only the variables available in the Dubai Land Department transactions data. Important factors such as property condition, renovations, building amenities, and broader economic indicators are not included. These missing variables can limit model accuracy, especially for luxury properties.

XII. Future Work

Future work can test stronger boosting models, add richer location features, and include market-level indicators. Another direction is training on earlier years and testing on later years to better simulate real use and check whether the model stays stable over time.

Acknowledgment

We would like to thank Professor Milan Dordevic (University of Wollongong Dubai, mdordevic@uowmail.edu.au) for supervision and guidance throughout this project. We would also like to thank Instructor Asma Damankesh (University of Wollongong Dubai, asma@uow.edu.au) for support and feedback during the course.

XIII. Conclusion

This paper applied machine learning to Dubai real estate sales for two tasks: predicting prices (regression) and classifying properties into price groups (classification). Random Forest achieved the best regression results (R^2 = 0.894950), improving performance from 0.7663 to 0.8950 (a 16.79% increase) and reducing prediction error compared to Linear Regression. For classification, Decision Tree outperformed Logistic Regression with 0.8843 accuracy/F1 compared to 0.8113, an improvement of about 7 percentage points.

Overall, tree-based models performed better for this dataset because they can capture nonlinear patterns and feature interactions that linear models miss. In practice, more accurate predictions can help users compare neighborhoods more fairly and estimate reasonable price ranges for similar properties, while the substantial error reduction observed with Random Forest can help investors and analysts reduce mispricing risk when evaluating large numbers of transactions.

References

[1] Investopedia Editors, "Hedonic pricing: Definition, uses, and real estate," Online. Available: https://www.investopedia.com/terms/h/hedonicpricing.asp, 2025, accessed: Feb. 27, 2026.

[2] scikit-learn Developers, "Randomforestregressor — scikit-learn documentation," Online. Available: https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html, 2026, accessed: Feb. 27, 2026.

[3] A. E. Balila *et al.*, "Comparative analysis of machine learning algorithms for predicting dubai property prices," *Frontiers in Applied Mathematics and Statistics*, 2024, accessed: Feb. 27, 2026.

[4] F. I. Al Marzooqi, "Predicting real estate prices using machine learning in abu dhabi," Online. Available: https://repository.cud.ac.ae/bitstreams/2f3331e7-ff49-4b37-a4c7-61bcc65316a0/download, 2024, accessed: Feb. 27, 2026.

[5] A. Alshamsi, "Prediction of dubai apartment prices using machine learning," Online. Available: https://repository.rit.edu/theses/11179/, 2022, accessed: Feb. 27, 2026.

[6] scikit-learn Developers, "scikit-learn: Machine learning in python (documentation)," Online. Available: https://scikit-learn.org/stable/, 2026, accessed: Feb. 27, 2026.

[7] ——, "Cross-validation: Evaluating estimator performance — scikit-learn user guide," Online. Available: https://scikit-learn.org/stable/modules/cross_validation.html, 2026, accessed: Feb. 27, 2026.

[8] ——, "Logisticregression — scikit-learn documentation," Online. Available: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html, 2026, accessed: Feb. 27, 2026.

[9] ——, "Decisiontreeclassifier — scikit-learn documentation," Online. Available: https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html, 2026, accessed: Feb. 27, 2026.

[10] Dubai Land Department, "Dubai land department transactions dataset," Online. Available: https://data.dubai/en/l/470061, 2026, accessed: Feb. 28, 2026.